UNIVERSITÄT BONN

@PuckerLab

Prof. Dr. Boas Pucker

Python - Artificial Intelligence

Availability of slides

- All materials are freely available (CC BY) - after the lectures:
  - GitHub: https://github.com/bpucker/PyBo

- Questions: Feel free to ask at any time

- Feedback, comments, or questions: pucker[a]uni-...

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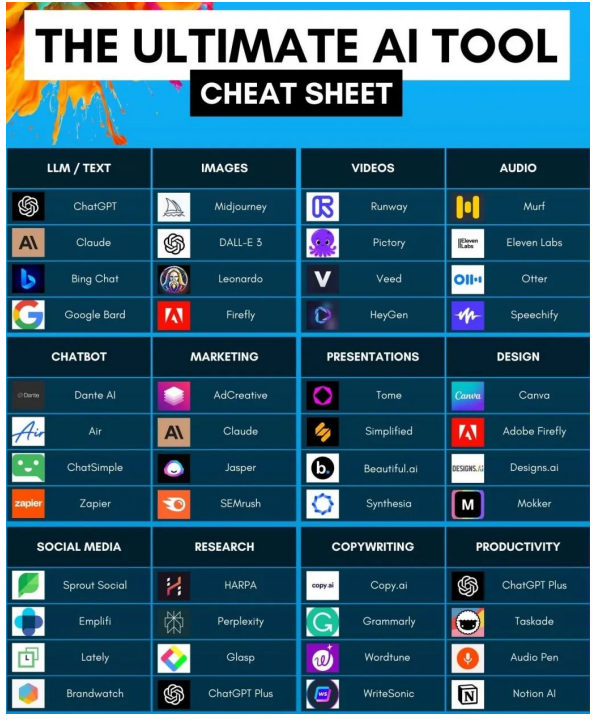
Introduction to Artificial Intelligence (AI)

- “simulation of human intelligence processes by machines, especially computer systems”
- Examples: Google searches, Netflix recommendations, self driving cars, ChatGPT3
- AI envisioned in 1956, but only took of after 2017
- LLM = Large Language Models

AI options



https://www.instagram.com/p/C067XZ5PKde/

Introduction to AI-Generated Code

- Definition: AI-powered tools and techniques for code generation.
- Significance: Transforming software development, automating repetitive tasks, enhancing productivity.
- Impact: Revolutionizing coding practices, enabling new possibilities in software engineering.

AI for Code Translation and Refactoring

- **Translation Between Script/Programming Languages**:
  - Applications: Porting projects, integrating libraries, interoperability.

- **Refactoring Tools**:
  - Features: suggesting improvements, optimizing performance.
  - Challenges: Maintaining functionality, preserving code quality, handling edge cases.

Code Generation with Generative Models

- **Generative Models**:
  - Definition: Models capable of generating new data samples.
  - Applications: Generating code snippets, functions, scripts.

- **Considerations**:
  - Ethical Implications: Ensuring code quality, security vulnerabilities.
  - Practical Challenges: Handling code complexity, maintaining readability.

Ethical and Practical Considerations

- **Risks & Limitations**:
  - Quality Assurance: Importance of human oversight, code reviews.
  - Security Concerns: Identifying and mitigating vulnerabilities, ensuring data privacy.

- **Best Practices**:
  - Responsible Use: Understanding AI limitations, continuous monitoring.
  - Collaboration: Balancing AI automation with human expertise, fostering teamwork.

ChatGPT3.5

You
How to generate a plot in python that visualizes a list of values?

ChatGPT
To generate a plot in Python that visualizes a list of values, the most commonly used library is `matplotlib`. Here's a step-by-step guide on how to do it:

1. **Install Matplotlib** (if you haven't already):

bash Copy code
```bash
pip install matplotlib
```

2. **Generate a Simple Plot**:
Here's a basic example to plot a list of values.

python Copy code
```python
import matplotlib.pyplot as plt

# Sample data
values = [1, 2, 3, 4, 5]

# Create a plot
plt.plot(values)

# Show the plot
plt.show()
```

3. **Customize the Plot**:
You can customize various aspects of the plot like adding titles, labels, changing colors, line styles, etc.

python Copy code
```python
import matplotlib.pyplot as plt

# Sample data
values = [1, 2, 3, 4, 5]
labels = ['A', 'B', 'C', 'D', 'E']

# Create a plot
plt.plot(labels, values, marker='o', linestyle='--', color='g', label=

# Add titles and labels
plt.title('Sample Plot')
plt.xlabel('Categories')
plt.ylabel('Values')

# Show legend
plt.legend()

# Show the plot
plt.show()
```


https://chat.openai.com

ChatGPT limitations

- V3 refuses to report scientific publications (used to make them up)
  - V4 is able to provide accurate references
- V3 refuses to report sequences
- Quality depends on input data and knowledge is outdated
- ChatGPT3.5 performs above average human performance in many cases
- ChatGPT4 was trained on AI generated code; potentially lower performance

Popular LLMs

- ChatGPT (OpenAI)
- Gemini (Google)
- Claude (Anthropic)
- Copilot (Microsoft)
- Perplexity
- Bing Chat (requires Microsoft Edge as web browser)

Exercises

- Repeat the exercises of all previous parts with AI support
- Take notes about interesting observations
- Is the AI solution better than your own solution?
- What are the advantages of using an AI?
- Are there disadvantages of the AI solution?

Time for questions!